

Module 07: TCP Error Control and Retransmission Mechanics

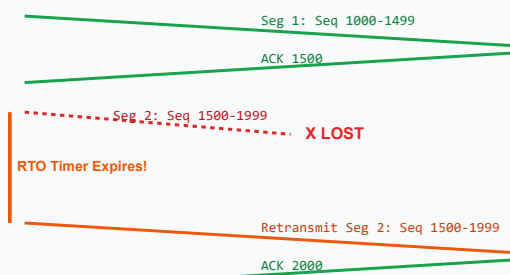
Unit 4: Transport Layer

Reliability & Retransmissions

Checksum Integrity, Cumulative ACKs, Jacobson's Dynamic RTO Mathematical Formula, Karn's Ambiguity Rule, and Three Duplicate ACKs Fast Retransmit.

TCP Error Control: RTO Calculation, Karn's Rule & Fast Retransmit

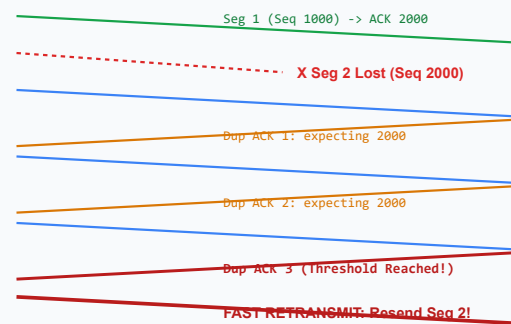
1. Lost Segment & RTO Timer Expiration



Jacobson's RTO Mathematical Formulas:

$\text{EstimatedRTT} = (1 - a) * \text{EstimatedRTT} + a * \text{SampleRTT} \quad (a = 1/8)$
 $\text{DevRTT} = (1 - b) * \text{DevRTT} + b * |\text{SampleRTT} - \text{EstimatedRTT}| \quad (b = 1/4)$
 $\text{RTO} = \text{EstimatedRTT} + 4 * \text{DevRTT}$
Karn's Rule: Do NOT measure SampleRTT for retransmitted segments!

2. Three Duplicate ACKs & Fast Retransmit



Why 3 Duplicate ACKs (Fast Retransmit)?

- Out-of-order delivery can generate 1 or 2 duplicate ACKs.
 - Receiving **3 Duplicate ACKs** guarantees segment loss!
 - Sender resends missing packet **before RTO timer expires**.
- Saves entire seconds of idle waiting time!**

1. Jacobson's Dynamic RTO Equations

$\text{EstimatedRTT} = 0.875 * \text{EstimatedRTT} + 0.125 * \text{SampleRTT}$

$\text{DevRTT} = 0.75 * \text{DevRTT} + 0.25 * |\text{SampleRTT} - \text{EstimatedRTT}|$

$\text{RTO} = \text{EstimatedRTT} + 4 * \text{DevRTT}$

Karn's Rule: Do NOT update RTT estimates for retransmitted segments!

2. Retransmission Triggers

Trigger Type	Detection Condition	Action Taken
RTO Expiration	No ACK arrives before RTO countdown hits 0	Slow retransmission; drops cwnd to 1 MSS
Fast Retransmit	Reception of 3 Duplicate ACKs	Instant retransmission without waiting for RTO!